

WATERING SYSTEM – USER MANUAL

This document explains system operation, TFT rows navigation, settings, and background AI behavior.

ROW 01 – TIME & DATE

Enter: short ENTER press when row 01 is selected.
UP/DOWN adjust values.
ENTER moves through fields (hour → minute → month → day → year).
Final ENTER saves and exits.

ROW 02 – MOIST / WET / DRY

Enter edit: hold ENTER for 5 seconds.
Blink indicates active field.
UP/DOWN adjust RAW values (step 10).
Short ENTER switches between WET and DRY.
Hold ENTER ~2 seconds to save and exit.
These values are user-locked (AI will not modify them).

ROW 03 – TARGET / DURATION

Enter: short ENTER.
TARGET – desired moisture (30–85%).
DURATION – watering time (3–10 minutes).
UP/DOWN adjust values.
ENTER switches between TARGET and DURATION, then exits.

ROW 04 – SCALE (SCL)

Enter: ENTER on row 04.
SCL = moisture scaling (60–100%).
UP/DOWN adjust by 1%.
ENTER saves and triggers immediate measurement.
Affects both display and watering logic.

CAL STATUS

CAL+ = calibration OK.
CAL~ = learning in progress.
CAL! = issue detected (sensor or range).

ROW 07 – SENSOR (SENSE)

Displays sensor status.

RAW value = direct reading.

OPEN = no soil contact.

SHORT = short circuit.

BACKGROUND AI (AL)

AI monitors moisture history, temperature, and soil behavior.

Calculates drying rate and predicts when moisture drops below TARGET.

Detects anomalies (jumps, stuck values, no recovery).

Does NOT modify user WET/DRY settings.

WHEN TO EXPECT ACCURATE RESULTS

Basic stabilization: after initial measurements.

AI accuracy: after 1–3 days of operation.

Optimal performance: after 3–5 days continuous use.